

Contact Information

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LinkedIn QR

Team

Jinho Hyun | Founder & CEO

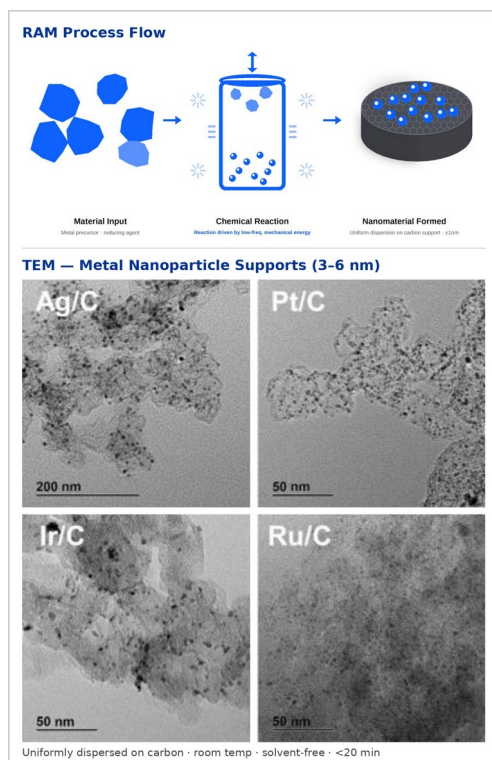
- Ph.D. in Engineering, GIST; 10 years of nanomaterial synthesis research
- Led national R&D projects totaling KRW 18.6B (NRF, Samsung SDI, etc.)
- Fuel cell / water electrolysis catalyst development; 4 SCIE/KCI papers

Chanho Pak | CTO

- Professor of Chemistry, GIST; 30 years of research
- Former Vice President, Samsung SDI
- Led pilot production line setup, process standardization & QA

R&D Team

- AI researcher (process modeling), material synthesis and electrochemical evaluation researchers joining 2026–27



Problem

- Each nanomaterial needs its own dedicated wet-synthesis line, driving high CAPEX and slow commercialization
- Yield, reproducibility and quality collapse during scale-up, so promising materials rarely leave the lab
- Wet synthesis relies on high heat, organic solvents and up to 48 hours per batch (~192 kgCO₂ per kg)

Solution

- A single RAM platform produces metal nanoparticles, nanoclusters, doped carbon, oxides, alloys, MOFs and MXenes
- Low-frequency resonant vibration replaces heat: 3 steps, 2 raw materials, room temperature, zero organic solvent

Key Features

- Up to 144× faster (≤20 min vs. 48 h), 1/72 power use and 1/16 CO₂ emissions per 1 kg batch
- Uniform 2–3 nm Pt/C (2.78 ± 0.33 nm); mass activity 0.450 A/mgPt, meeting the DOE target
- Manufacturing cost cut up to 80% (non-precious) / 41% (precious); 99% reproducibility, <1% loss

Business Model

- B2B direct sales of catalyst materials (Pt/C, IrO₂, doped carbon, metal oxides) to hydrogen, battery and chemical makers
- ODC custom development (NRE fees) converting into long-term supply agreements
- Global technology licensing (upfront fee + royalty) and carbon-credit-linked ESG supply partnerships

Business Snapshot

- Founded Aug 2026; incubated at GIST Startup Center; selected for 11 government programs (KRW 230M non-dilutive secured)
- 4 PoCs in progress with 3 industrial partners (Companies C, K, O) since Dec 2025; NDA and PoC underway with Company G
- 1 patent filed (Sep 2026); 12 core process/material patents planned incl. PCT
- Awards: 2026 Climate-Energy-Environment Startup Competition (Excellence), 2026 KAIST Climate Tech Audition (Innovation)
- Raising KRW 100M pre-seed to build a 1.5 kg/batch pilot line; 2030 revenue target KRW 30B

Target Market

- Fuel cell & water electrolysis (hydrogen), battery and chemical/environmental catalyst manufacturers
- Nano & high-performance catalyst market: \$7.1B (2025) → \$13.9B (2035), CAGR 7.0%

Seeking Opportunities

- **PoC & sample-evaluation partners among fuel cell, electrolysis, battery and chemical catalyst makers**
- **Seed investors and global licensing / distribution partners**